Chapter 13

Campbell Biology, AP Edition, 12e

Two children have similar features to their parents, including bone structure, eye shape, and skin color.

Chapter 13

Meiosis and Sexual Life Cycles

Figure 13.1

The family members shown in this photo have some similar features. Offspring resemble their parents more than they do unrelated individuals.

What biological mechanisms account for the resemblance between offspring and their parents?

Meiosis is a special type of cell division that produces cells with half the chromosomes of the parent cell. It occurs only in specialized cells, such as the cells of the testes and ovaries in humans. Humans have 45 chromosomes. Here, each parent cell shows only one pair of homologous chromosomes (homologs). (They carry genes for the same characters.) Chromosomes are duplicated before meiosis begins. Meiosis 1: Homologs are separated. Meiosis 2: sister chromatids are separated. The chromosome in each sperm is one of a pair of homologs, and the chromosome in each egg is one of a pair of homologs. Fertilization unites sperm and egg, re-establishing pairs of homologous chromosomes, with both paternal and maternal genes.

Key Concepts

- **13.1Offspring acquire genes from parents by inheriting chromosomes****
- **13.2Fertilization and meiosis alternate in sexual life cycles****
- **13.3Meiosis reduces the number of chromosome sets from diploid to haploid****
- **13.4Genetic variation produced in sexual life cycles contributes to evolution****

Advanced placement edition

Meiosis followed by fertilization ensures genetic diversity in sexually reproducing organisms **(BIG IDEA 3)** and provides genetic variation that plays a role in natural selection **(BIG IDEA 1)**. This cellular process is

driven by the interaction of subcellular components **(BIG IDEA 4)** and uses energy that is required for the growth and reproduction of living systems **(BIG IDEA 2)**.

Study Tip

Make a visual study guide: Figure 13.1 presents the events of meiosis as a simplified diagram. As you learn more about meiosis, draw a detailed diagram of the stages of meiosis, starting with the sketch below. Include explanatory labels. Compare your visual study guides of mitosis (from Chapter 12) and meiosis, listing similarities and differences.

A simplified diagram of meiosis is captioned, Cell ready to start meiosis (chromosomes have been duplicated).

Get Ready for This Chapter

In this chapter, you will learn about the mechanisms of meiotic cell division and the three types of sexual life cycles, with their alternation of meiosis and fertilization. Before beginning this chapter, you should be able to do the following:

- Describe what happens to a chromosome before a cell divides (see Concept 12.1).
- Explain the progression of the cell cycle (see Concept 12.2 and Figure 12.6).
- Describe the process of mitosis (see Concept 12.2 and Figure 12.7).
- Explain how the amount of DNA changes during the cell cycle (see Concept 12.2).

The following questions provide a quick check of your basic knowledge in these areas.

1. What must happen to a chromosome before a cell starts mitosis?

```
    The single DNA molecule in the chromosome must be
replicated.
```

That's correct.

Correct! If you had trouble with this question, review the following material: Before a cell starts mitosis, its chromosomes must be duplicated. This includes replicating the single DNA molecule of a chromosome, resulting in two sister chromatids of a duplicated chromosome that are attached along their lengths. The chromosomes condense during prophase of mitosis.

```
        Both molecules of DNA in the chromosome must be
replicated.
```

That's incorrect. Try again.

Incorrect. Review the following material: Before a cell starts mitosis, its chromosomes must be duplicated. This includes replicating the single DNA molecule of a chromosome, resulting in two sister chromatids of a duplicated chromosome that are attached along their lengths. The chromosomes condense during prophase of mitosis.

```
        The chromatin of the chromosome must be completely
condensed.
```

That's incorrect. Try again.

Incorrect. Review the following material: Before a cell starts mitosis, its chromosomes must be duplicated. This includes replicating the single DNA molecule of a chromosome, resulting in two sister chromatids of a duplicated chromosome that are attached along their lengths. The chromosomes condense during prophase of mitosis.

```
        The two chromatids of the chromosome must become
attached to each other.
```

That's incorrect. Try again.

Incorrect. Review the following material: Before a cell starts mitosis, its chromosomes must be duplicated. This includes replicating the single DNA molecule of a chromosome, resulting in two sister chromatids of a duplicated chromosome that are attached along their lengths. The chromosomes condense during prophase of mitosis.

2. Which of the following phases make up the stages of mitosis?

```
     Interphase
```

That's incorrect. Try again.

Incorrect. Review the following material: Mitosis is conventionally broken down into five stages: prophase, prometaphase, metaphase, anaphase, and telophase. Overlapping with the latter stages of mitosis, cytokinesis completes the mitotic phase M phase but is separate from mitosis. Interphase, accounting for the remainder of the cell cycle, can be divided into three subphases: G1 phase, S phase, and G2 phase.

G1, S, and G2

That's incorrect. Try again.

Incorrect. Review the following material: Mitosis is conventionally broken down into five stages: prophase, prometaphase, metaphase, anaphase, and telophase. Overlapping with the latter stages of mitosis, cytokinesis completes the mitotic phase M phase but is separate from mitosis. Interphase, accounting for the remainder of the cell cycle, can be divided into three subphases: G1 phase, S phase, and G2 phase.

```
        Prophase, prometaphase, metaphase, anaphase, telophase,
and cytokinesis
```

That's incorrect. Try again.

Incorrect. Review the following material: Mitosis is conventionally broken down into five stages: prophase, prometaphase, metaphase, anaphase, and telophase. Overlapping with the latter stages of mitosis, cytokinesis completes the mitotic phase M phase but is separate from mitosis. Interphase, accounting for the remainder of the cell cycle, can be divided into three subphases: G1 phase, S phase, and G2 phase.

```
        Prophase, prometaphase, metaphase, anaphase, and
telophase
```

That's correct.

Correct! If you had trouble with this question, review the following material: Mitosis is conventionally broken down into five stages: prophase, prometaphase, metaphase, anaphase, and telophase. Overlapping with the latter stages of mitosis, cytokinesis completes the mitotic phase M phase but is separate from mitosis. Interphase, accounting for the remainder of the cell cycle, can be divided into three subphases: G1 phase, S phase, and G2 phase.

3. Which of the following events characterizes metaphase of mitosis?

```
     alignment of the chromosomes at the equator
```

That's correct.

Correct! If you had trouble with this question, review the following material: At metaphase, the chromosomes' centromeres line up at the metaphase plate, a plane that is equidistant between the spindle's two poles. DNA condenses into discrete chromosomes during prophase, and centrosomes move away from each other during prometaphase. Sister chromatids are separated during anaphase.

```
condensing of the DNA into discrete chromosomes
```

That's incorrect. Try again.

Incorrect. Review the following material: At metaphase, the chromosomes' centromeres line up at the metaphase plate, a plane that is equidistant between the spindle's two poles. DNA condenses into discrete chromosomes during prophase, and centrosomes move away from each other during prometaphase. Sister chromatids are separated during anaphase.

```
movement of the chromosomes toward the equator
```

That's incorrect. Try again.

Incorrect. Review the following material: At metaphase, the chromosomes' centromeres line up at the metaphase plate, a plane that is equidistant between the spindle's two poles. DNA condenses into discrete chromosomes during prophase, and centrosomes move away from each other during prometaphase. Sister chromatids are separated during anaphase.

```
separation of sister chromatids
```

That's incorrect. Try again.

Incorrect. Review the following material: At metaphase, the chromosomes' centromeres line up at the metaphase plate, a plane that is equidistant between the spindle's two poles. DNA condenses into discrete chromosomes during prophase, and centrosomes move away from each other during prometaphase. Sister chromatids are separated during anaphase.

4. Measurements of the amount of DNA per nucleus were taken on a large number of cells from a growing fungus. The measured DNA levels ranged from 3 to 6 picograms (pg) per nucleus. During the entirety of which stage of the cell cycle did the nucleus contain 6 pg of DNA?

S

That's incorrect. Try again.

Incorrect. Review the following material: A cell in G1 that has 3 pg of DNA duplicates its DNA entirely during S phase, a process that results in 6 pg of DNA. During the entirety of G2, the cell has 6 pg of DNA as the cell continues to grow and complete preparations for cell division. The cell then divides during M phase, resulting in two daughter cells that have 3 pg of DNA each, identical to that of the parent cell in G1.

G1

That's incorrect. Try again.

Incorrect. Review the following material: A cell in G1 that has 3 pg of DNA duplicates its DNA entirely during S phase, a process that results in 6 pg of DNA. During the entirety of G2, the cell has 6 pg of DNA as the cell continues to grow and complete preparations for cell division. The cell then divides during M phase, resulting in two daughter cells that have 3 pg of DNA each, identical to that of the parent cell in G1.

M

That's incorrect. Try again.

Incorrect. Review the following material: A cell in G1 that has 3 pg of DNA duplicates its DNA entirely during S phase, a process that results in 6 pg of DNA. During the entirety of G2, the cell has 6 pg of DNA as the cell continues to grow and complete preparations for cell division. The cell then divides during M phase, resulting in two daughter cells that have 3 pg of DNA each, identical to that of the parent cell in G1.

G2

That's correct.

Correct!If you had trouble with this question, review the following material: A cell in G1 that has 3 pg of DNA duplicates its DNA entirely during S phase, a process that results in 6 pg of DNA. During the entirety of G2, the cell has 6 pg of DNA as the cell continues to grow and complete preparations for cell division. The cell then divides during M phase, resulting in two daughter cells that have 3 pg of DNA each, identical to that of the parent cell in G1.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 13.1Compare and contrast asexual and sexual reproduction with respect to inheritance of chromosomes by offspring.

- 13.2Explain the alternation of fertilization and meiosis in different types of sexual life cycles, using the terms haploid, diploid and zygote.
- 13.3Describe the stages of meiosis, explaining how the process reduces the number of chromosome sets.
- 13.4Identify the ways in which sexual life cycles generate genetic variation that contributes to evolution.

Concept 13.1: Offspring acquire genes from parents by inheriting chromosomes

Have you ever heard someone say they have their mother's nose or their father's eyes? The transmission of traits from one generation to the next is called inheritance, or **heredity** (from the Latin *heres*, heir). At the same time, sons and daughters are not identical copies of either parent or of their siblings. Along with inherited similarity, there is also **variation**. The study of both heredity and inherited variation is called **genetics**.

Inheritance of Genes

Parents endow their offspring with coded information in the form of hereditary units called **genes**. The genes we inherit are our genetic link to our parents, and they account for family resemblances such as shared eye color or freckles. Our genes program specific traits that emerge as we develop from fertilized eggs into adults.

The genetic program is written in the language of DNA, the polymer of four different nucleotides you learned about in Concepts 1.1 and 5.5. Inherited information is passed on in the form of each gene's specific sequence of DNA nucleotides, much as printed information is communicated in the form of meaningful sequences of letters. In both cases, the language is symbolic. Just as your brain translates the word *apple* into a mental image of the fruit, cells translate genes into freckles and other features. Most genes program cells to synthesize specific enzymes and other proteins, whose cumulative action produces an organism's inherited traits. The programming of these traits in the form of DNA is one of the unifying themes of biology.

The transmission of hereditary traits has its molecular basis in the replication of DNA, which produces copies of genes that can be passed from parents to offspring. In animals and plants, reproductive cells called ****gametes** are the vehicles that transmit genes from one generation to the next. During fertilization, male and female gametes (sperm and eggs) unite, passing on genes of both parents to their offspring.

Sperm fertilizing egg. Sperm is much smaller than egg.

Sperm and egg

Except for small amounts of DNA in mitochondria and chloroplasts, the DNA of a eukaryotic cell is packaged into chromosomes within the nucleus. Every species has a characteristic number of chromosomes. For example, humans have 46 chromosomes in their ****__somatic cells__—all cells of the body except the gametes and their precursors. Each chromosome consists of a single long DNA molecule, elaborately coiled in association with various proteins. One chromosome includes several hundred to a few thousand genes, each gene being a precise sequence of nucleotides along the DNA molecule. A gene's specific location along the length of a chromosome is called the gene's **locus** (plural, *loci*; from the Latin, meaning "place"). Our genetic endowment (our genome) consists of the genes and other DNA that make up the chromosomes we inherited from our parents.

Comparison of Asexual and Sexual Reproduction

Only organisms that reproduce asexually have offspring that are exact genetic copies of themselves. In **asexual reproduction**, a single individual (like a yeast cell or an amoeba; see Figure 12.2a) is the sole parent and passes copies of all its genes to its offspring without the fusion of gametes. For example, single-celled eukaryotic organisms can reproduce asexually by mitotic cell division, in which DNA is copied and allocated equally to two daughter cells. The genomes of the offspring are virtually exact copies of the parent's genome. Some multicellular organisms are also capable of reproducing asexually **(Figure 13.2)**. Because the cells of the offspring arise via mitosis in the parent, the offspring is usually genetically identical to its parent. An individual that reproduces asexually gives rise to a **clone**, an individual or group of individuals that are genetically identical to the parent. Genetic differences occasionally arise in asexually reproducing organisms as a result of changes in the DNA called mutations, which we will discuss in Concept 17.5.

Figure 13.2

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Asexual reproduction in two multicellular organisms.
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(a) Hydra: Parent has a tubular body with round projections at one end. Bud resembles a small arm coming from main parent body. Scale: 0.5 millimeters. (b) Redwoods: Circle of five redwood trees clustered around a stump.

(a) *This relatively simple animal, a hydra, reproduces by budding. The bud, a localized mass of mitotically dividing cells, develops into a small hydra, which detaches from the parent (LM).* ***(b)*** *All the trees in this circle of redwoods arose asexually from a single parent tree, whose stump is in the center of the circle.*

Video

Hydra Budding

image

Play

image

Play

Animation

Asexual Reproduction

image

Play

image

Play

In **sexual reproduction**, two parents give rise to offspring that have unique combinations of genes inherited from the two parents. In contrast to a clone, offspring of sexual reproduction vary genetically from their siblings and both parents: They are variations on a common theme of family resemblance, not exact replicas. Genetic variation like that shown in Figure 13.1 is an important consequence of sexual reproduction. What mechanisms generate this genetic variation? The key is the behavior of chromosomes during the sexual life cycle.

Concept Check 13.1

- **MAKE CONNECTIONS** Using your knowledge of gene expression in a cell, explain what causes the traits of parents (such as hair color) to show up in their offspring. (See Concept 5.5.)
- Describe how an asexually reproducing eukaryotic organism produces offspring that are genetically identical to each other and to their parents.
- **WHAT IF?** A horticulturalist breeds orchids, trying to obtain a plant with a unique combination of desirable traits. After many years, she finally succeeds, and wants to produce more plants like this one. Discuss whether she should crossbreed it with another plant or cause it to undergo asexual reproduction (forming a clone), and why.

*For suggested answers, see*Appendix A.**

Concept 13.2: Fertilization and meiosis alternate in sexual life cycles

A **life cycle** is the generation-to-generation sequence of stages in the reproductive history of an organism, from conception to production of its own offspring. In this section, we use humans as an example to track the behavior of chromosomes through the sexual life cycle. We begin by considering the chromosome count in human somatic cells and gametes. We will then explore how the behavior of chromosomes relates to the human life cycle and other types of sexual life cycles.

Sets of Chromosomes in Human Cells

In humans, each somatic cell has 46 chromosomes. Before mitosis begins, the chromosomes are duplicated. During mitosis, the chromosomes become condensed enough to be visible under a light microscope. At this point, they can be distinguished from one another by their size, the position of their centromeres, and the pattern of colored bands produced by certain chromatin-binding stains.

Careful examination of a micrograph of the 46 human chromosomes from a single cell in mitosis reveals that there are two chromosomes of each of 23 types. This becomes clear when images of the chromosomes are arranged in pairs, starting with the longest chromosomes. The resulting ordered display is called a **karyotype(Figure 13.3)**. The two chromosomes of a pair have the same length, centromere position, and staining pattern: These are called **homologous chromosomes** (or **homologs**). Both chromosomes of each pair carry genes controlling the same inherited characters. For example, if a gene for eye color is situated at a particular locus on a certain chromosome, then its homologous chromosome (its homolog) will also have a version of the eye-color gene at the equivalent locus.

Figure 13.3

Research Method: Preparing a Karyotype

Application

A karyotype is a display of condensed chromosomes arranged in pairs. Karyotyping can be used to screen for defective chromosomes or abnormal numbers of chromosomes associated with certain congenital disorders, such as Down syndrome.

A technician looks through a microscope.

Technique

Karyotypes are prepared from isolated somatic cells, which are treated with a drug to stimulate mitosis and then grown in culture for several days. Cells arrested when the chromosomes are most highly condensed—at metaphase—are stained and then viewed with a microscope equipped with a digital camera. An image of the chromosomes is displayed on a computer monitor, and digital software is used to arrange them in pairs according to their appearance.

A human karyotype.

Human karyotype is an image of 23 pairs of homologous chromosomes, labeled 1 through 22 for somatic cells. Chromosomes clearly vary in length and color banding patterns, but chromosomes in each homologous pair show similar length and color banding patterns. X and Y chromosomes are visible after chromosome pair 22, indicating that the karyotype is from a human male. The X chromosome is larger than the Y chromosome and contains different color banding patterns.

The image zooms in on metaphase homologous pair 1, with a diagram of this pair at the side of the karyotype. The diagram shows this pair of homologous chromosomes in sister chromatid form, indicating that they have been duplicated and have similar centromere positions.

Results

This karyotype shows the chromosomes from a human male (as seen by the presence of the XY chromosome pair), colored to emphasize their chromosome banding patterns. The size of the chromosome, position of the centromere, and pattern of stained bands help identify specific chromosomes. Although difficult to discern in the karyotype, each metaphase chromosome consists of two closely attached sister chromatids (see the diagram of the first pair of homologous duplicated chromosomes).

MAKE CONNECTIONS *Explain why each chromosome in the karyotype consists of two sister chromatids. (See**Figures 12.6**and*12.7.)*

*For suggested answer, see*Appendix A.*

The two chromosomes referred to as X and Y are an important exception to the general pattern of homologous chromosomes in human somatic cells. Typically, human females have a homologous pair of X chromosomes (XX), while males have one X and one Y chromosome (XY; see Figure 13.3). Only small parts of the X and Y are homologous. Most of the genes carried on the X chromosome do not have counterparts on the tiny Y, and the Y chromosome has genes lacking on the X. Due to their role in sex determination, the X and Y chromosomes are called **sex chromosomes**. The other chromosomes are called **autosomes**.

The occurrence of pairs of homologous chromosomes in each human somatic cell is a consequence of our sexual origins. We inherit one chromosome of a pair from each parent. Thus, the 46 chromosomes in our somatic cells are actually two sets of 23 chromosomes—a maternal set (from our female parent) and a paternal set (from our male parent). The number of chromosomes in a single set is represented by n*.* Any cell with two chromosome sets is called a **diploid cell** and has a diploid number of chromosomes, abbreviated 2n. For humans, the diploid number is 46(2n=46), the number of chromosomes in our somatic cells. In a cell in which DNA synthesis has occurred, all the chromosomes are duplicated, and therefore each consists of two identical sister chromatids, associated closely at the centromere and along the arms. (Even though the chromosomes are duplicated, we still say the cell is diploid, or 2n. This is because it has only two sets of information regardless of the number of chromatids, which are merely copies of the information in one set.) **Figure 13.4** helps clarify the various terms that we use to describe duplicated chromosomes in a diploid cell.

Figure 13.4

Describing chromosomes.

Cell (2 n equals to 6) with maternal set of chromosomes (n equals to 3) and paternal set of chromosomes (n equals to 3). Each of the six chromosomes is duplicated into sister chromatids.

Homologous chromosome sets are distinguished by color with blue homologs representing the paternal set of chromosomes and red homologs representing the maternal. Each homologous pair contains one maternal and one paternal homolog. Homologous chromosome pair #1 is long, homologous chromosome pair #2 is medium length, and homologous chromosome pair #3 is short. Here all six chromosomes are in duplicated sister chromatid form; sister chromatids are identified as being the same color and attached to each other at the centromere. Homologous chromosomes contain one blue set of sister chromatids and one red set for each of the long, medium and short pairs. Within a homologous pair (same length maternal and paternal chromosome sets), one blue sister chromatid would be a nonsister chromatid to a red chromatid. Nonsister chromatids are not attached to each other.

A cell from an organism with a diploid number of 62n=6 is depicted here following chromosome duplication and condensation. Each of the six duplicated chromosomes consists of two sister chromatids associated closely along their lengths. Each homologous pair is composed of one chromosome from the maternal set (red) and one from the paternal set (blue). Each set is made up of three chromosomes in this example (long, medium, and short). Together, one maternal and one paternal chromatid in a pair of homologous chromosomes are called nonsister chromatids.

VISUAL SKILLS *How many sets of chromosomes are present in this diagram? How many pairs of homologous chromosomes are present? If the phases of mitosis in* ***Figure 12.7*** *were redrawn using the color scheme shown in this figure, describe how the six chromosomes would be colored.*

*For suggested answer, see*Appendix A.**

BioFlix® Animation

Chromosomes

image

Play

image

Play

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Unlike somatic cells, gametes contain a single set of chromosomes. Such cells are called **haploid cells**, and each has a haploid number of chromosomes (n). For humans, the haploid number is 23(n=23). The set of 23 consists of the 22 autosomes plus a single sex chromosome. An unfertilized egg contains an X chromosome; a sperm contains either an X or a Y chromosome.

Each sexually reproducing species has a characteristic diploid and haploid number. For example, the fruit fly *Drosophila melanogaster* has a diploid number (2n) of 8 and a haploid number (n) of 4, while for dogs, 2n is 78 and n is 39. The chromosome number generally does not correlate with the size or complexity of a species' genome; it simply reflects how many linear pieces of DNA make up the genome, which is a function of the evolutionary history of that species (see Concept 21.5). Now let's consider chromosome behavior during sexual life cycles. We'll use the human life cycle as an example.

Behavior of Chromosome Sets in the Human Life Cycle

The human life cycle begins when a haploid sperm from the father fuses with a haploid egg from the mother **(Figure 13.5)**. This union of gametes, culminating in fusion of their nuclei, is called ********fertilization**. The resulting fertilized egg, or **zygote**, is diploid because it contains two haploid sets of chromosomes bearing genes representing the maternal and paternal family lines. As a human develops into a sexually mature adult, mitosis of the zygote and its descendant cells generates all the somatic cells of the body. Both chromosome sets in the zygote and all the genes they carry are passed with precision to the somatic cells.

Figure 13.5

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The human life cycle.
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The human life cycle with haploid stages existing after meiosis and until fertilization, and diploid stages following fertilization and continuing until meiosis in the offspring (production of the gametes).

This figure introduces a color code that will be used for other life cycles later in this book. The aqua arrows identify haploid stages of a life cycle, and the tan arrows identify diploid stages.

Arrows depict the human life cycle containing both haploid and diploid stages.

Multicellular diploid adults (a woman and a man) with 2 n equals to 46. Tan (diploid) arrows indicate meiosis occurs in diploid cells in the woman's ovary or the man's testes, with aqua arrows following meiosis indicating that a haploid stage has begun. The gametes resulting from meiosis (eggs for the woman and sperm for the man) are both haploid (n equals to 23). Aqua (haploid) arrows flow to fertilization, where the gametes meet to form a diploid zygote (2 n equals to 46). Tan (diploid) arrows flow from the diploid zygote through the processes of mitosis and development back to the multicellular diploid adult stage (2 n equals to 46), at which point the cycle begins again.

In each generation, the number of chromosome sets per cell is halved during meiosis but doubles at fertilization. For humans, the number of chromosomes in a haploid cell is 23 consisting of one set n=23; the number of chromosomes in the diploid zygote and all somatic cells arising from it is 46, consisting of two sets 2n=46.

Animation

The Human Life Cycle

image

Play

image

Play

The only cells of the human body not produced by mitosis are the gametes, which develop from specialized cells called *germ cells* in the gonads—ovaries in females and testes in males (see Figure 13.5). Imagine what would happen if human gametes were made by mitosis: They would be diploid like the somatic cells. At the next round of fertilization, when two gametes fused, the normal chromosome number of 46 would double to 92, and each subsequent generation would double the number of chromosomes yet again. This does not happen, however, because in sexually reproducing organisms, gamete formation involves a type of cell division called **meiosis**. This type of cell division reduces the number of sets of chromosomes from two in the parent cell to one in each gamete, counterbalancing the doubling that occurs at fertilization. As a result of meiosis, each human sperm and egg is haploid (n=23). Fertilization restores the diploid condition by

combining two sets of chromosomes, and the human life cycle is repeated, generation after generation (see Figure 13.5).

The steps of the human life cycle are typical of many sexually reproducing animals. Indeed, fertilization and meiosis are also the hallmarks of sexual reproduction in plants, fungi, and protists just as in animals. Fertilization and meiosis alternate in sexual life cycles, maintaining a constant number of chromosomes in a species from one generation to the next.

The Variety of Sexual Life Cycles

Although the alternation of meiosis and fertilization is common to all organisms that reproduce sexually, the timing of these two events in the life cycle varies, depending on the species. These variations can be grouped into three main types of life cycles. In the type that occurs in humans and most other animals, gametes are the only haploid cells **(Figure 13.6a)**. Meiosis occurs in germ cells during the production of gametes, which undergo no further cell division prior to fertilization. After fertilization, the diploid zygote divides by mitosis, producing a multicellular organism that is diploid.

Figure 13.6

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Three types of sexual life cycles.
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Three types of sexual life cycles.

The common feature of all three cycles is the alternation of meiosis and fertilization, key events that contribute to genetic variation among offspring. The cycles differ in the timing of these two key events. (Small circles are cells; large circles are organisms.)

Types of cycles are as follows.

- Animals. Meiosis produces haploid gametes from cells in the diploid multicellular organism (mouse). Fertilization results from the fusion of haploid gametes; the resulting diploid zygote develops via mitosis. Diploid (2 n) multicellular organism (mouse) flows via diploid (tan) arrows to meiosis; haploid (aqua) arrows begin and flow to gametes (n). Fusion with another haploid gamete results in fertilization; diploid (tan) arrows flow to the zygote (2 n). Diploid arrows flow back to the multicellular mouse via mitosis and the cycle continues.
- b. Plants and some algae. The life cycle includes multicellular diploid sporophyte alternating with multicellular haploid gametophyte; haploid spores result from meiosis; gametes produced via mitosis. Fusion of gametes equals fertilization. Cells in diploid (2 n) multicellular organisms (sporophyte) undergo meiosis, flowing by haploid arrows to produce the spores (n). Spores flow via

haploid arrows by mitosis to form a haploid multicellular organism called the gametophyte. Haploid arrows flow from the gametophyte to produce the haploid gametes via mitosis. Fusion with another haploid gamete results in fertilization, after which diploid arrows flow to the diploid zygote (2 n). Diploid arrows flow from the zygote back to the sporophyte via mitosis and the cycle begins again.

- c. Most fungi and some protists. Meiosis produces haploid cells that develop via mitosis into haploid organism. From here, gametes produced via mitosis fuse via fertilization to produce diploid zygote. A haploid unicellular or multicellular organism produces haploid gametes via mitosis; fusion with another haploid gamete flows via haploid arrows to fertilization. Diploid arrows flow to diploid zygote (2 n), from which haploid cells are produced via meiosis. Some haploid cells flow via haploid arrows and the process of mitosis back to the haploid unicellular organism and the cycle begins again.

VISUAL SKILLS *For each type of life cycle, indicate whether haploid cells undergo mitosis, and if they do, describe the cells that are formed.*

*For suggested answer, see*Appendix A.**

Plants and some species of algae exhibit a second type of life cycle called **alternation of generations(Figure 13.6b)**. This type includes both diploid and haploid stages that are multicellular. The multicellular diploid stage is called the *sporophyte*. Meiosis in the sporophyte produces haploid cells called *spores*. Unlike a gamete, a haploid spore doesn't fuse with another cell but divides mitotically, generating a multicellular haploid stage called the *gametophyte*. Cells of the gametophyte give rise to gametes by mitosis. Fusion of two haploid gametes at fertilization results in a diploid zygote, which develops into the next sporophyte generation. Therefore, in this type of life cycle, the sporophyte generation produces a gametophyte as its offspring, and the gametophyte generation produces a sporophyte. The name *alternation of generations* fits well for this type of life cycle.

A third type of life cycle occurs in most fungi and some protists, including some algae **(Figure 13.6c)**. After gametes fuse and form a diploid zygote, meiosis occurs without a multicellular diploid offspring developing. Meiosis produces not gametes but haploid cells that then divide by mitosis and give rise to either unicellular descendants or a haploid multicellular adult organism. Subsequently, the haploid organism carries out further mitoses, producing the cells that develop into gametes. The only diploid stage found in these species is the single-celled zygote.

Note that *either* haploid or diploid cells can divide by mitosis, depending on the type of life cycle. Only diploid cells, however, can undergo meiosis: Haploid cells can't because they already have a single set of chromosomes that cannot be further reduced. Though the three types of sexual life cycles differ in the timing of meiosis and fertilization, they share a fundamental result: genetic variation among offspring.

Concept Check 13.2

- **MAKE CONNECTIONS** In Figure 13.4, how many DNA molecules (double helices) are present (see Figure 12.5)? What is the haploid number of this cell? Is a set of chromosomes haploid or diploid?
- **VISUAL SKILLS** In the karyotype in Figure 13.3, how many pairs of chromosomes are present? How many sets?
- Using shoes as an analogy for chromosomes, how would you describe the collection of "shoes" in human diploid and haploid cells?
- **WHAT IF?** A certain eukaryote lives as a unicellular organism, but during environmental stress, it produces gametes. The gametes fuse, and the resulting zygote undergoes meiosis, generating new single cells. What type of organism could this be?

*For suggested answers, see*Appendix A.**

Concept 13.3: Meiosis reduces the number of chromosome sets from diploid to haploid

Several steps of meiosis closely resemble corresponding steps in mitosis. Meiosis, like mitosis, is preceded by interphase, which includes S phase (the duplication of chromosomes). However, this is followed by not one but two consecutive cell divisions, called **meiosis I** and **meiosis II**. These two divisions result in four daughter cells (rather than the two daughter cells of mitosis), each with only half as many chromosomes as the parent cell—one set, rather than two.

The Stages of Meiosis

The overview of meiosis in **Figure 13.7** shows, for a single pair of homologous chromosomes in a diploid cell, that both members of the pair are duplicated and the copies sorted into four haploid daughter cells. Recall that sister chromatids are two copies of *one* chromosome, closely associated all along their lengths; this association is called *sister chromatid cohesion*. Together, the sister chromatids make up one duplicated chromosome (see Figure 13.4). In contrast, the two chromosomes of a homologous pair are individual chromosomes that were inherited from each parent. Homologs appear alike in the microscope, but they may have different versions of genes at corresponding loci; each version is called an *allele* of that gene (see Figure 14.4). For example, one chromosome might have an allele for freckles, but the homologous chromosome may have an allele for the absence of freckles at the same locus. Homologs are not associated with each other in any obvious way except during meiosis.

Figure 13.7

```
Overview of meiosis: how meiosis reduces chromosome number.
```

Diploid cell with one pair homologous chromosomes goes through Interphase, Meiosis 1 and Meiosis 2. Four haploid daughter cells result, each with one chromosome (two with paternal homologs; 2 maternal).

After the chromosomes duplicate in interphase, the diploid cell divides twice, yielding four haploid daughter cells. This overview tracks just one pair of homologous chromosomes, which for the sake of simplicity are drawn in the condensed state throughout. (They would not normally be condensed during interphase.)

Interphase: Diploid cell drawn with one pair of homologous chromosomes: one maternal (red) and one paternal (blue) homolog. Arrows indicate duplication of chromosomes, with the result being a diploid cell with duplicated chromosomes: maternal and paternal homologs both in sister chromatid form (sister chromatids attached at centromere).

Meiosis 1: Homologous chromosomes separate. The result is two haploid cells, each with a duplicated chromosome (paternal sister chromatids in one cell, maternal sister chromatids in the other cell. Sister chromatids remain attached at the centromere).

Meiosis 2: Sister chromatids separate. The result is four haploid daughter cells: 2 from the paternal chromatids, each cell now containing a single paternal chromosome, and 2 from the maternal chromatids, each cell now containing a single maternal chromosome. The four haploid daughter cells contain unduplicated chromosomes.

DRAW IT *Redraw the cells in this figure using a simple double helix to represent each DNA molecule.*

*For suggested answer, see*Appendix A.**

Animation

Overview of Meiosis

image

image

**

Figure 13.8 describes in detail the stages of the two divisions of meiosis for an animal cell whose diploid number is 6. Study this figure thoroughly before going on.

Figure 13.8 Interactive Visual Activity

Exploring Meiosis in an Animal Cell

Interact with this figure in the Figure 13.8 Interactive Visual Activity: Exploring Meiosis in an Animal Cell.

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<p>iframes are not supported by your browser.</p>
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Figure 13.8

```
Exploring Meiosis in an Animal Cell
```

MEIOSIS 2: Separates sister chromatids and is made up of Prophase 2, Metaphase 2, Anaphase 2, Telophase 2 and Cytokinesis.

During another round of cell division, the sister chromatids finally separate in Anaphase 2; four haploid daughter cells result, containing unduplicated chromosomes.

Prophase 2:

- A spindle apparatus forms.
- In late prophase 2 (not shown here), chromosomes, each still composed of two chromatids associated at the centromere, is moved by microtubules toward the metaphase 2 plate.

Metaphase 2:

- The chromosomes are positioned at the metaphase plate as in mitosis.
- Because of crossing over in meiosis 1, the two sister chromatids of each chromosome are not genetically identical.
- The kinetochores of sister chromatids are attached to microtubules extending from opposite poles.

Anaphase 2:

- Breakdown of proteins holding the sister chromatids together at the centromere allows the chromatids to separate. The chromatids move toward opposite poles as individual chromosomes.

Telophase 2 and cytokinesis:

- Nuclei form, the chromosomes begin decondensing, and cytokinesis occurs.
- The meiotic division of one parent cell produces four daughter cells, each with a haploid set of (unduplicated) chromosomes.

- The four daughter cells are genetically distinct from one another and from the parent cell.

MAKE CONNECTIONS *Look at**Figure 12.7**and imagine the two daughter cells undergoing another round of mitosis, yielding four cells. Compare the number of chromosomes in each of those four cells, after mitosis, with the number in each cell in*Figure 13.8*, after meiosis. Explain how the process of meiosis results in this difference, even though meiosis also includes two cell divisions.**

*For suggested answer, see*Appendix A.**

Video

Meiosis I in Sperm Formation

image

Play

image

Play

BioFlix® Animation

Meiosis

image

Play

image

**

Animation

Meiosis

image

Play

image

Play

Crossing Over and Synapsis During Prophase I

Prophase I of meiosis is a very busy time. The prophase I cell shown in Figure 13.8 is at a point fairly late in prophase I, when pairing of homologous chromosomes, crossing over, and chromosome condensation have already taken place. The sequence of events leading up to that point is shown in more detail in **Figure 13.9**.

Figure 13.9

```
Crossing over and synapsis in prophase I: a closer look.
```

A diagram of crossing over and synapsis in a series of four steps.

Steps are as follows.

- After interphase, the chromosomes have been duplicated, and sister chromatids are held together by proteins called cohesins (purple). Each pair of homologs is associated along their length. The DNA molecules of two non-sister chromatids are broken at precisely corresponding points. The chromatin of the chromosomes starts to condense.
- A zipper-like protein complex, the synaptonemal complex (green), begins to form, attaching one homolog to the other. The chromatin continues to condense.
- a. The synaptonemal complex is fully formed; the two homologs are said to be in synapsis. During synapsis, the DNA breaks are closed up when each broken end is joined to the corresponding segment of the non-sister chromatid, producing crossovers. After the synaptonemal complex disassembles, the homologs move slightly apart from each other but remain attached because of sister chromatid cohesion, even though some of the DNA may no longer be attached to its original chromosome. The points of attachment where crossovers have occurred show up as chiasmata. The chromosomes continue to condense as they move toward the metaphase plate.

VISUAL SKILLS *Label the four DNA breaks in step 2.*

*For suggested answer, see*Appendix A.**

After interphase, the chromosomes have been duplicated and the sister chromatids are held together by proteins called *cohesins*. **STEP 1** Early in prophase I, the two members of a homologous pair associate loosely along their length. Each gene on one homolog is aligned precisely with the corresponding allele of that gene on the other homolog. The DNA of two nonsister chromatids—one maternal and one paternal—is broken by specific proteins at precisely matching points. **STEP 2** Next, the formation of a zipper-like structure called the **synaptonemal complex** holds one homolog tightly to the other. **STEP 3** During this association, called **synapsis**, the DNA breaks are closed up so that each broken end is joined to the

corresponding segment of the *nonsister* chromatid. Thus, a paternal chromatid is joined to a piece of maternal chromatid beyond the crossover point, and vice versa.

STEP 4 These points where crossing over has just occurred become visible as chiasmata (singular, **chiasmata**) after the synaptonemal complex disassembles and the homologs move slightly apart from each other. The homologs remain attached because sister chromatids are still held together by sister chromatid cohesion, even though some of the DNA may no longer be attached to its original chromosomal DNA. At least one crossover per chromosome must occur in order for the homologous pair to stay together as it moves to the metaphase I plate, for reasons that will be explained shortly.

A Comparison of Mitosis and Meiosis

Figure 13.10 summarizes the key differences between meiosis and mitosis in diploid cells. Meiosis produces four cells and reduces the number of chromosome sets from two (diploid) to one (haploid), whereas mitosis produces two cells and conserves the number of chromosome sets. Meiosis produces cells that differ genetically from the parent cell and from each other, whereas mitosis produces daughter cells that are genetically identical to their parent cell and to each other.

Figure 13.10

```
A comparison of mitosis and meiosis.
```

Comparison of Mitosis and Meiosis during Prophase, Metaphase, Anaphase and Telophase.

Two tables are given.

The first table is as follows.

Mitosis	Meiosis
Prophase: Parent cell (before chromosome duplication) is shown (2 n equals to 6). Chromosome duplication is initiated and duplicated chromosomes are formed (two sister chromatids)	(Meiosis 1) Prophase 1: Parent cell (before chromosome duplication) is shown (2 n equals to 6). Chromosome duplication is initiated and homologous chromosome pairs are formed which are held together by chiasma (site of crossing over) and sister chromatid cohesion.
Metaphase: Individual chromosomes line up at the metaphase plate.	Metaphase 1: Pairs of homologous chromosomes line up at the metaphase plate.

Anaphase, Telophase: Sister chromatids separate during anaphase and 2 daughter cells of mitosis are formed (2 n)	Anaphase 1, Telophase 1: Homologs separate during anaphase 1; sister chromatids remain attached at centromere. Daughter cells of meiosis 1 are formed (Haploid n equals to 3)
Anaphase, Telophase: Sister chromatids separate during anaphase and 2 daughter cells of mitosis are formed (2 n)	Sister chromatids separate during anaphase 2 creating daughter cells (n) of meiosis 2

The second table is as follows.

Summary
Property
DNA replication
Number of divisions
Synapsis of homologous chromosomes
Number of daughter cells and genetic composition
Role in animals, fungi, and plants

DRAW IT *Could any other combinations of chromosomes be generated during meiosis II from the specific cells shown in telophase I? Explain. (Hint: Draw the cells as they would appear in metaphase II.)*

*For suggested answer, see*Appendix A.**

Figure 13.10 Walkthrough

A comparison of mitosis and meiosis

Watch an explanation of this figure in the Figure 13.10 Walkthrough: A comparison of mitosis and meiosis.

image

Play

image

Three events unique to meiosis occur during meiosis I:

- **Synapsis and crossing over.** During prophase I, duplicated homologs pair up and crossing over occurs as described previously and in Figure 13.9. Synapsis and crossing over do not occur during prophase of mitosis.
- **Alignment of homologous pairs at the metaphase plate.** At metaphase I of meiosis, pairs of homologs are positioned at the metaphase plate, rather than individual chromosomes, as in metaphase of mitosis.
- **Separation of homologs.** At anaphase I of meiosis, the duplicated chromosomes of each homologous pair move toward opposite poles, but the sister chromatids of each duplicated chromosome remain attached. In anaphase of mitosis, by contrast, sister chromatids separate.

Sister chromatids stay together due to sister chromatid cohesion, mediated by cohesin proteins. In mitosis, this attachment lasts until the end of metaphase, when enzymes cleave the cohesins, freeing the sister chromatids to move to opposite poles of the cell. In meiosis, sister chromatid cohesion is released in two steps: cohesion of *arms* at the start of anaphase I and cohesion of *centromeres* at anaphase II. In metaphase I, the two homologs of each pair are held together because there is still cohesion between sister chromatid arms in regions beyond points of crossing over, where stretches of sister chromatids now belong to different chromosomes. The combination of crossing over and sister chromatid cohesion along the arms results in the formation of a chiasma. Chiasmata hold homologs together as the spindle forms for the first meiotic division. At the onset of anaphase I, the release of cohesion along sister chromatid arms allows homologs to separate. At anaphase II, the release of sister chromatid cohesion at the centromeres allows the sister chromatids to separate. Thus, sister chromatid cohesion and crossing over, acting together, play an essential role in the lining up of chromosomes by homologous pairs at metaphase I.

Meiosis I reduces the number of chromosome sets: from two (diploid) to one (haploid). During the second meiotic division, sister chromatids separate, producing four haploid daughter cells. The mechanisms for separating sister chromatids in meiosis II and mitosis are virtually identical. The molecular basis of chromosome behavior during meiosis continues to be a focus of intense research. In the **Scientific Skills Exercise**, you can work with data that track the amount of DNA in cells as they progress through meiosis.

Scientific Skills Exercise

Making a Line Graph and Converting Between Units of Data

AP Science Practices 2, 4, 5

How Does DNA Content Change as Budding Yeast Cells Proceed Through Meiosis? When nutrients are low, cells of the budding yeast (*Saccharomyces cerevisiae*) exit the mitotic cell cycle and enter

meiosis. In this exercise, you will track the DNA content of a population of yeast cells as they progress through meiosis.

Budding yeast cells look similar to eggs.

Budding yeast cells

How the Experiment Was Done Researchers grew a culture of yeast cells in a nutrient-rich medium and then transferred the cells to a nutrient-poor medium to induce meiosis. At different times after induction, the DNA content per cell was measured in a sample of the cells, and the average DNA content per cell was recorded in femtograms (fg; 1 femtogram=1×10−15 gram).

Data from the Experiment

A table lists the time after induction of meiosis (hour) and the average amount of D N A per cell (femtograms).

The table is as follows.

Time after induction of meiosis in hours	Average amount of D N A per cell in femtograms
0	24
1	24
2	40
3	47
4	47.5
5	48
6	48
7	47.5
7.5	25
8	24
9	23.5

9.5	14
10	13
11	12.5
12	12
13	12.5
14	12

Further Reading G. Simchen, Commitment to meiosis: What determines the mode of division in budding yeast? *BioEssays* 31:169–177 (2009).

INTERPRET THE DATA

- First, set up your graph.
- Place the labels for the independent and dependent variables on the appropriate axes, followed by units of measurement in parentheses. Explain your choices.
- Add tick marks and values for each axis. Explain your choices. (For additional information about graphs, see the Scientific Skills Review in Appendix D.)
- Because the variable on the x-axis varies continuously, it makes sense to plot the data on a line graph (if you are using Excel, choose "Scatter").
- Plot each data point from the table onto the graph.
- Connect the data points with line segments.
- Most of the yeast cells in the culture were in G1 of the cell cycle before being moved to the nutrient-poor medium.
- How many femtograms of DNA are there in each yeast cell in G1? Estimate this value from the data in your graph.
- How many femtograms of DNA should be present in each cell in G2? (See Concept 12.2 and Figure 12.6.) At the end of meiosis I (MI)? At the end of meiosis II (MII)? (See Figure 13.7.)
- Using these values as a guideline, distinguish the different phases by inserting vertical dashed lines in the graph between phases and label each phase (G1, S, G2, MI, MII). You can figure out where to

put the dividing lines based on what you know about the DNA content of each phase (see Figure 13.7).

- Think carefully about the point where the line at the highest value begins to slope downward. What specific point of meiosis does this “corner” represent? What stage(s) correspond(s) to the downward sloping line?
- Given the fact that 1 fg of DNA=9.78×105 base pairs (on average), you can convert the amount of DNA per cell to the length of DNA in numbers of base pairs.
- Calculate the number of base pairs of DNA in the haploid yeast genome. Express your answer in millions of base pairs (Mb), a standard unit for expressing genome size. Show your work.
- How many base pairs per minute were synthesized during the S phase of these yeast cells?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Concept Check 13.3

- **MAKE CONNECTIONS** Compare the chromosomes in a cell at metaphase of mitosis with those in a cell at metaphase II. (See Figures 12.7 and 13.8.)
- **WHAT IF?** After the synaptonemal complex disappears, how would any pair of homologous chromosomes be associated if crossing over did not occur? What effect might this have on gamete formation?

*For suggested answers, see*Appendix A.**

Concept 13.4: Genetic variation produced in sexual life cycles contributes to evolution

How do we account for the genetic variation of the family members in Figure 13.1? As you will learn more about in later chapters, mutations are the original source of genetic diversity. These changes in an organism’s DNA create the different versions of genes, known as alleles. Once these differences arise, reshuffling of the alleles during sexual reproduction produces the variation that results in each member of a sexually reproducing population having a unique combination of traits.

Origins of Genetic Variation Among Offspring

In species that reproduce sexually, the behavior of chromosomes during meiosis and fertilization is responsible for most of the variation that arises in each generation. Three mechanisms contribute to the genetic variation arising from sexual reproduction: independent assortment of chromosomes, crossing over, and random fertilization.

Independent Assortment of Chromosomes

One aspect of sexual reproduction that generates genetic variation is the random orientation of pairs of homologous chromosomes at metaphase of meiosis I. At metaphase I, the homologous pairs, each consisting of one maternal and one paternal chromosome, are situated at the metaphase plate. (Note that the terms *maternal* and *paternal* refer, respectively, to whether the chromosome in question was contributed by the mother or the father of the individual whose cells are undergoing meiosis.) Each pair may orient with either its maternal or paternal homolog closer to a given pole—its orientation is as random as the flip of a coin. Thus, there is a 50% chance that a particular daughter cell of meiosis I will get the maternal chromosome of a certain homologous pair and a 50% chance that it will get the paternal chromosome.

Because each pair of homologous chromosomes is positioned independently of the other pairs at metaphase I, the first meiotic division results in each pair sorting its maternal and paternal homologs into daughter cells independently of every other pair. This is called *independent assortment*. Each daughter cell represents one outcome of all possible combinations of maternal and paternal chromosomes. As shown in **Figure 13.11**, the number of combinations possible for daughter cells formed by meiosis of a diploid cell with two pairs of homologous chromosomes (n=2) is four: two possible arrangements for the first pair times two possible arrangements for the second pair. Note that only two of the four combinations of daughter cells shown in the figure would result from meiosis of a *single* diploid cell because a single parent cell would have one or the other possible chromosomal arrangement at metaphase I, but not both. However, the population of daughter cells resulting from meiosis of a large number of diploid cells contains all four types in approximately equal numbers. In the case of n=3, eight combinations (2×2×2=23) of chromosomes are possible for daughter cells. More generally, the number of possible combinations when chromosomes sort independently during meiosis is 2n, where n is the haploid number of the species.

Figure 13.11

```
    The independent assortment of homologous chromosomes in
meiosis.
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Two equally probable arrangements of chromosomes at metaphase 1 are shown with the resulting daughter cells (Possibility 1 and Possibility 2).

Two equally probable arrangements of chromosomes at metaphase 1 are shown with the resulting daughter cells (Possibility 1 and Possibility 2). There are four chromosomes in two homologous pairs (a long homologous pair and a short homologous pair) present in the parent cell.

- Possibility 1 shows the homologous pairs oriented at Metaphase 1 as follows: duplicated long paternal chromosome oriented with duplicated short paternal chromosome and duplicated long maternal chromosome oriented with duplicated short maternal chromosome.
- Following Meiosis 1 (homologous pairs separate), two haploid daughter cells are shown in Metaphase 2 (sister chromatids remain attached at the centromeres) with one cell containing two duplicated paternal chromosomes (one long and one short) and one cell with two duplicated maternal chromosomes (one long and one short).
- Following Meiosis 2 (sister chromatids separate) the four haploid daughter cells that result contain long paternal and short paternal chromosomes (2) and long maternal and short maternal chromosomes (2).
- Possibility 2 shows the homologous pairs in Metaphase 1 oriented as follows: duplicated long paternal chromosome oriented with duplicated short maternal chromosome and duplicated long maternal chromosome oriented with duplicated short paternal chromosome.
- After Meiosis 1 (homologous pairs separate and daughter cells are haploid) one daughter cell contains duplicated long paternal and duplicated short maternal chromosomes, and the other daughter cell contains the duplicated long maternal and short paternal chromosomes. Both daughter cells shown in Metaphase 2.
- After Meiosis 2 (sister chromatids separate) the four haploid daughter cells contain a long paternal and short maternal chromosome (2) and a long maternal and short paternal chromosome (2).

Animation

Genetic Variation from Independent Assortment of Chromosomes

image

Play

image

Play

**

In the case of humans (n=23), the number of possible combinations of maternal and paternal chromosomes in the resulting gametes is 223, or about 8.4 million. Each gamete that you produce in your lifetime contains one of roughly 8.4 million possible combinations of chromosomes. This is an underestimate, because it doesn’t take into account crossing over, which we’ll consider next.

Crossing Over

As a consequence of the independent assortment of chromosomes during meiosis, each of us produces a collection of gametes differing greatly in their combinations of the chromosomes we inherited from our two parents. Figure 13.11 suggests that each chromosome in a gamete is exclusively maternal or paternal in origin. In fact, this is *not* the case, because crossing over produces **recombinant chromosomes**, individual chromosomes that carry genes (DNA) from two different parents **(Figure 13.12)**. In meiosis in humans, an average of one to three crossover events occurs per chromosome pair, depending on the size of the chromosomes and the position of their centromeres.

Figure 13.12

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The results of crossing over during meiosis.
```

The result of crossing over during meiosis is shown including Prophase 1, Anaphase 1, Anaphase 2 and the creation of daughter cells. Inset TEM of Chiasma is shown.

The result of crossing over during meiosis is shown. At Prophase 1 homologous pairs line up and non-sister chromatids are held together during synapsis.

Step 1: In prophase 1, synapsis and crossing over occur; then homologs move apart slightly.

Step 2: Chiasmata and attachments between sister chromatids hold homologs together; they move to the metaphase 1 plate. Chiasma, the site of crossing over, is shown both in a diagram as well as T E M view. The centromere is also shown.

Step 3: Breakdown of proteins holding sister chromatid arms together allows homologs with recombinant chromatids to separate. The results of crossing over are drawn here by showing small bits of maternal chromosome (red) now on one paternal chromosome and small bits of paternal chromosome (blue) now on a maternal chromosome, creating recombinant chromosomes.

Anaphase 2 I is shown with 2 cells with 2 chromosomes each – one cell with a paternal and recombinant chromosome and one cell with a recombinant and maternal chromosome. Following Meiosis 2 the four daughter cells are shown – one with a paternal chromosome, one with a maternal chromosome, and two recombinant chromosomes.

Animation

Genetic Variation from Crossing Over

image

Play

image

Play

As you learned in Figure 13.9, crossing over produces chromosomes with new combinations of maternal and paternal alleles. At metaphase II, chromosomes that contain one or more recombinant chromatids can be oriented in two alternative, nonequivalent ways with respect to other chromosomes because their sister chromatids are no longer identical (see Figure 13.12). The different possible arrangements of nonidentical sister chromatids during meiosis II further increase the number of genetic types of daughter cells that can result from meiosis.

You'll learn more about crossing over in Concept 15.3. The important point for now is that crossing over, by combining DNA inherited from two parents into a single chromosome, is an important source of genetic variation in sexual life cycles.

Random Fertilization

The random nature of fertilization adds to the genetic variation arising from meiosis. In humans, each male and female gamete represents one of about 8.4 million (223) possible chromosome combinations due to independent assortment. The fusion of a male gamete with a female gamete during fertilization will produce a zygote with any of about 70 trillion (223×223) diploid combinations. If we factor in the variation brought about by crossing over, the number of possibilities is truly astronomical. You really *are* unique.

Animation

Genetic Variation from Random Fertilization

image

Play

image

Play

The Evolutionary Significance of Genetic Variation Within Populations

EVOLUTION Now that you've learned how new combinations of genes arise among offspring in a sexually reproducing population, how does the genetic variation in a population relate to evolution? Darwin

recognized that a population evolves through the differential reproductive success of its variant members. On average, those individuals best suited to the local environment leave the most offspring, thereby transmitting their genes. Thus, natural selection results in the accumulation of genetic variations favored in a certain environment. As the environment changes, the population may survive if, in each generation, at least some of its members can cope effectively with the new conditions. Mutations are the original source of different alleles, which are then mixed and matched during meiosis. New and different combinations of alleles may work better than those that previously prevailed.

In a stable environment, though, sexual reproduction seems as if it would be less advantageous than asexual reproduction, which ensures perpetuation of successful combinations of alleles. Furthermore, sexual reproduction is more expensive energetically than asexual reproduction. In spite of these apparent disadvantages, sexual reproduction is almost universal among animals. Why is this?

The ability of sexual reproduction to generate genetic diversity is the most commonly proposed explanation for the evolutionary persistence of this process. However, consider the unusual case of the bdelloid rotifer **(Figure 13.13)**. It appears that this group may not have reproduced sexually for more than 50 million years of their evolutionary history, a model that has been supported by recent analysis of the genetic sequences in its genome. Does this mean that genetic diversity is not advantageous in this species? It turns out that bdelloid rotifers are an exception to the “rule” that sex alone generates genetic diversity: Bdelloids have mechanisms other than sexual reproduction for generating genetic diversity. For example, they live in environments that can dry up for long periods of time, during which they can enter a state of suspended animation. In this state, their cell membranes may crack in places, allowing entry of DNA from other rotifer species and even from more distantly related species. Evidence suggests that this foreign DNA can become incorporated into the genome of the bdelloid, leading to increased genetic diversity. In fact, the genomic analysis shows that bdelloid rotifers pick up non-bdelloid DNA at a much higher rate than most other species pick up foreign DNA. The conclusion that bdelloid rotifers have developed other ways of achieving genetic diversity supports the idea that genetic diversity is advantageous, but points out as well that sexual reproduction is not the only way of generating such diversity.

Figure 13.13

A bdelloid rotifer, an animal that reproduces only asexually.

A bdelloid rotifer is a translucent animal on a 200 micrometer scale.

In this chapter, we have seen how sexual reproduction greatly increases the genetic variation present in a population. Although Darwin realized that heritable variation is what makes evolution possible, he could not explain why offspring resemble—but are not identical to—their parents. Ironically, Gregor Mendel, a contemporary of Darwin, published a theory of inheritance that helps explain genetic variation, but his discoveries had no impact on biologists until 1900, more than 15 years after Darwin (1809–1882) and

Mendel (1822–1884) had died. In the next chapter, you'll learn how Mendel discovered the basic rules governing the inheritance of specific traits.

Concept Check 13.4

- What is the original source of variation among the different alleles of a gene?
- The diploid number for fruit flies is 8, and the diploid number for grasshoppers is 46. If no crossing over took place, would the genetic variation among offspring from a given pair of parents be greater in fruit flies or grasshoppers? Explain.
- **WHAT IF?** If maternal and paternal chromatids have the same two alleles for every gene, will crossing over lead to genetic variation?

*For suggested answers, see*Appendix A.**